

Results Guide for Simaan and Paz

This document explains the exact analysis behind the scientific report. Figure and table numbers match the report. Use it to check the model's unit of analysis, the direction of each metric, and which conclusions are justified before presenting the work.

The sentence to remember: spline+PCA reconstructs observed shape better; the CVAE predicts timing and generated distributions better; n=8 is the stronger generator, while n=3 is the requested compact fingerprint. Explicit conditioning and cognitive strategy labels are not yet validated.

The project question

We are not asking whether one trial identifies a person with no prior data. We enroll an unseen participant from a context half of their trials, obtain one fingerprint center, and ask whether that fingerprint can generate the distribution of the participant's disjoint query trials. This is participant-level adaptation followed by held-out within-participant prediction.

What the CVAE sees

Encoder input: 100 x 2 filtered position samples plus task metadata. It does not receive participant identity, physical initiation time, physical movement time, or trial number. Decoder outputs: a phase-normalized x-y trajectory plus initiation and movement time. The fingerprint is the mean posterior coordinate over the participant's context trials. Averaging is meaningful within one trained VAE coordinate system, but latent axes cannot be compared directly across independently trained seeds without alignment.



Four outer participant folds; each participant tested once; neural seeds 42/43/44

Figure 1. The test participant is unseen during model fitting, but their context trials are intentionally used for enrollment. Their query trials remain untouched until evaluation.

Metric reference

Metric	Formula / comparison	Better	What it measures
Trajectory MSE	mean squared x-y error over 100 phases	lower	posterior-mean reconstruction of an observed trial
Timing MAE	mean predicted - observed , milliseconds	lower	typical absolute timing error; primary timing endpoint
R ²	$1 - \frac{\sum(y - \hat{y})^2}{\sum(y - \bar{y})^2}$	higher	variance explained; can be negative and is outlier-sensitive
KS statistic	maximum separation between two empirical CDFs	lower	one feature's generated-versus-query distribution mismatch
MMD	kernel distance between multivariate samples	lower	joint distribution mismatch after feature scaling
Energy distance	distance-based multivariate discrepancy	lower	joint mismatch; raw values are scale/outlier sensitive
Balanced accuracy	mean recall across seven enrolled subjects	higher	closed-set fingerprint enrollment, not zero-shot identification
TV / JSD	categorical distribution distance	lower	component-count distribution mismatch
Wilcoxon + BH-FDR	paired participant ranks + multiple-test correction	p < .05	whether a model contrast is consistent across participants

The inference unit

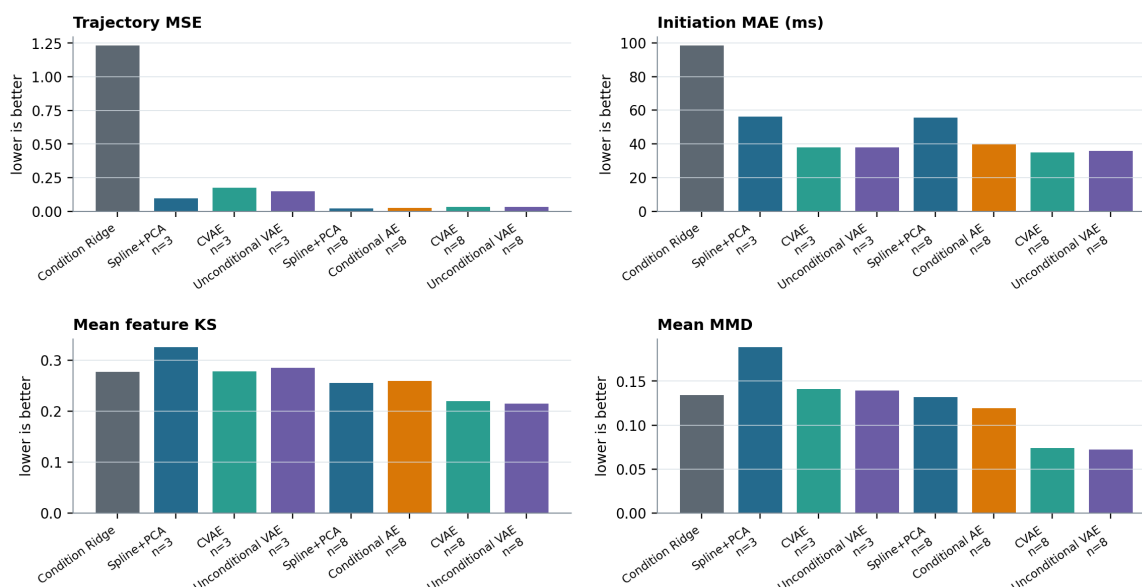
Neural seeds are repeated measurements of the same participant split. We first average seeds within each held-out participant, then run paired tests across 28 participants. Treating 3 seeds x 6 condition strata x 28 participants as independent would artificially inflate the sample size; that earlier aggregation was corrected before these reports.

Why MAE is primary for timing

Most timing predictions are close, but a few log-space decoder outputs extrapolate to impossible durations. MAE, median error, and the 95th percentile describe typical performance more honestly than a single pooled R². We still retain the unclipped R² and maximum error in raw outputs so failure cases are visible.

Reading Figure 2 and Table 1

Figure 2. Out-of-fold benchmark endpoints (28 held-out participants)



Each panel answers a different question. Trajectory MSE uses an observed test trial as encoder input, so it measures compression/reconstruction. KS and MMD use only a context fingerprint plus sampled latent noise and compare generated samples with separate query trials, so they measure distribution generation. A model can win one and lose the other without contradiction.

Model	n	Traj. MSE	Init. MAE ms	Move MAE ms	Mean KS	MMD	Enroll. acc.
Condition Ridge	-	1.234	98.5	72.7	0.277	0.134	-
Spline+PCA	3	0.097	56.2	51.2	0.325	0.189	36.4%
CVAE	3	0.177	37.8	43.4	0.278	0.141	40.9%
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.140	40.0%
Spline+PCA	8	0.023	55.6	50.2	0.255	0.132	48.6%
Conditional AE	8	0.026	39.9	42.7	0.259	0.119	46.5%
CVAE	8	0.032	34.8	44.2	0.219	0.074	50.7%
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.072	50.8%

What the rows say

Spline+PCA n=8 has the lowest trajectory MSE (0.023 versus CVAE 0.032). CVAE n=8 has better initiation MAE (34.8 versus 55.6 ms), movement MAE (44.2 versus 50.2 ms), mean KS (0.219 versus 0.255), and MMD (0.074 versus 0.132). Therefore 'CVAE is better' is too broad; 'CVAE is better for timing and generated distributions, while spline+PCA is better for deterministic shape reconstruction' is accurate.

What makes the comparison scientific

Table 2 compares the two methods participant by participant after seed averaging. 'CVAE better 23/28' means the CVAE had lower error for 23 of 28 held-out participants. The Wilcoxon test uses the signed ranks of all 28 paired differences; BH-FDR adjusts p-values across the family of model/metric comparisons.

Endpoint	CVAE	Spline+PCA	CVAE better	BH-FDR p
Trajectory MSE	0.177	0.097	6/28	3.3e-05
Initiation MAE	37.8	56.2	27/28	1.2e-07
Movement MAE	43.4	51.2	24/28	9.9e-05
Mean KS	0.278	0.325	26/28	1.6e-06
MMD	0.141	0.189	24/28	5.8e-05

Table 2a. CVAE $n=3$ versus spline+PCA $n=3$.

Endpoint	CVAE	Spline+PCA	CVAE better	BH-FDR p
Trajectory MSE	0.032	0.023	4/28	8.3e-04
Initiation MAE	34.8	55.6	27/28	1.2e-07
Movement MAE	44.2	50.2	23/28	0.0014
Mean KS	0.219	0.255	23/28	1.1e-04
MMD	0.074	0.132	26/28	3.1e-06

Table 2b. CVAE $n=8$ versus spline+PCA $n=8$.

Why the condition-only baseline matters

Without this baseline, good generated distributions might simply reflect the six task strata and population variability. At $n=3$, CVAE clearly improves reconstruction and timing but not mean KS/MMD over condition-only Ridge. At $n=8$, it improves all primary endpoints for most participants. This is the strongest evidence that the higher-capacity fingerprint carries useful participant information beyond task metadata.

What not to say

Do not say that $n=3$ fully predicts each person's distributions, that the CVAE is universally superior, or that negative R^2 means every timing prediction is bad. Use the endpoint-specific statements above.

Why $n=3$ and $n=8$ tell different parts of the story

Figure 3. CVAE capacity and fingerprint enrollment

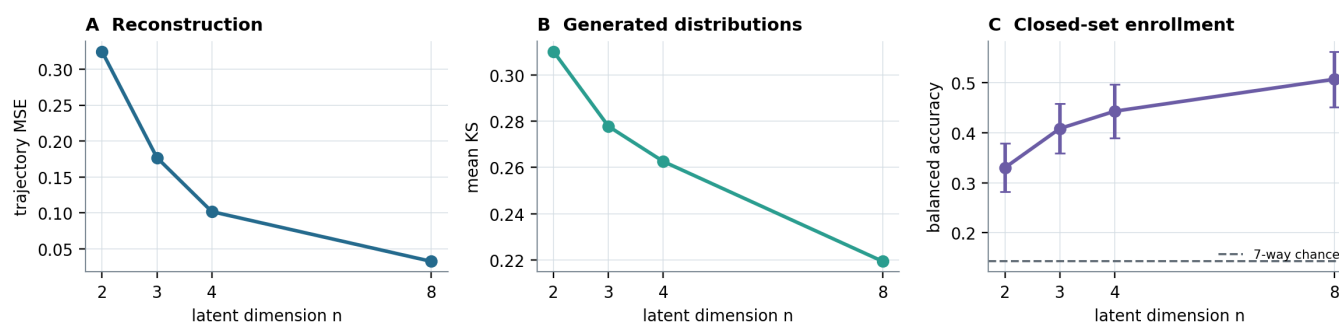


Figure 3A-B. More latent capacity lowers reconstruction and distribution error. **Figure 3C.** Enrollment also improves descriptively. This does not prove that every new coordinate has a separate human meaning; a VAE may rotate or distribute information across coordinates.

$n=3$

This matches Jason's desired 2-3 controls and is easy to visualize. It beats spline+PCA for timing and distribution fidelity, but it does not beat the condition-only baseline on the main distribution metrics. Treat it as the compact interpretation model, not the strongest final generator.

$n=8$

This model has enough capacity to improve generated distributions beyond the task-only baseline and gives the best CVAE fidelity. It is harder to interpret directly. Its result tells us the architecture can represent useful participant information, while $n=3$ may be too restrictive for complete fidelity.

Why two latent coordinates may correlate

The KL term encourages a standard-normal aggregate prior but does not guarantee independent semantic factors. Coordinate correlations within one fit can arise from the data manifold and from imperfect posterior factorization. Across seeds, signs and axis order can change. This is why the dashboard's heatmap is a run-specific exploratory explanation, not a cross-seed causal map of what each slider controls.

Did the task condition help?

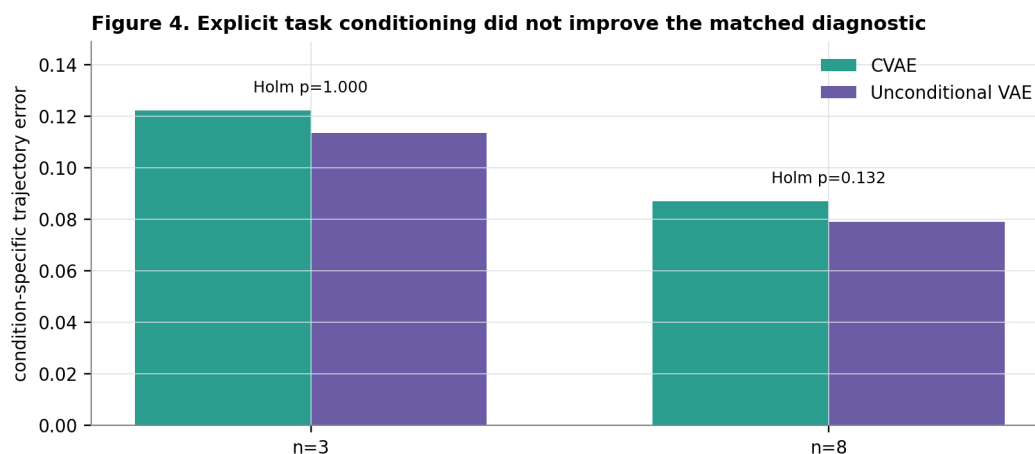


Figure 4. The CVAE receives task condition and the unconditional VAE does not. We hold the fingerprint and random latent draws fixed, change only the condition, and compare the resulting six sp x side strata with observed strata. If conditioning gave reliable participant-specific control, CVAE errors should be consistently lower.

They are not significantly lower: $n=3$ Holm $p=1.000$ and $n=8$ $p=0.132$ for the trajectory diagnostic, and none of 44 feature tests survives BH-FDR. Therefore the dashboard can expose the controls because the architecture uses them, but we cannot claim that moving a speed or side control reproduces the correct participant-specific effect.

Correct answer if asked: conditioning was motivated by task structure, but the matched unconditional ablation shows that its incremental value is not established in this dataset.

Why we keep the CVAE at all

The research interface requested by Jason is conditional: choose a fingerprint and task setting, then sample plausible movement distributions. The CVAE is the direct architecture for that interface. The ablation result means its conditioning claims must stay provisional, not that the model or experiment is invalid.

What a minimum-jerk component is

A component is one smooth 2-D velocity pulse described by onset time, duration, lateral displacement, and forward displacement. Several components can overlap so their summed velocity approximates a recorded movement. Component count is chosen by us after fitting 1, 2, 3, and 4 candidate models; Jason's repository does not choose the winning count automatically.

Figure 5. Minimum-jerk component analysis is model-order sensitive

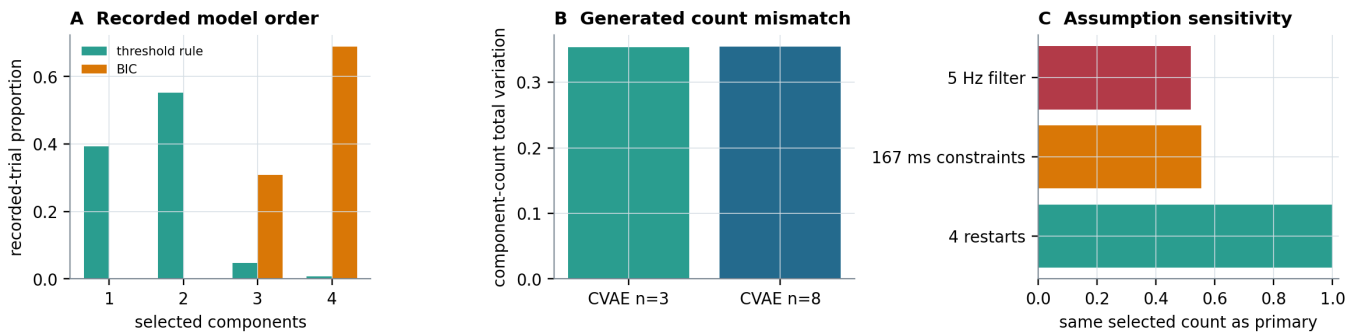


Figure 5A. Threshold selection and BIC produce very different recorded count distributions. **5B.** Total variation around 0.35 means the generated categorical count distribution still differs substantially from recorded query trials. **5C.** Extra optimizer restarts do not change the primary count, but filter and temporal constraints change almost half of counts.

Can count be called strategy?

Not yet. A one-component movement can be compatible with a wait-then-go interpretation and multiple components can be compatible with correction, but the decomposition is kinematic. The sensitivity result shows that count also depends on analysis settings. We need Jason to approve the filter, 167 ms question, and model-order criterion before mapping count to a cognitive strategy.

n=3 versus n=8

n=8 trends better for several continuous component parameters, but none of nine paired contrasts survives BH-FDR. It does not improve component-count TV or JSD. Therefore minimum-jerk analysis does not provide evidence for choosing n=8 over n=3.

Representative outputs are illustrations, not the test

Figure A1. Held-out reconstruction and context-fingerprint generation (fold 0, seed 42)

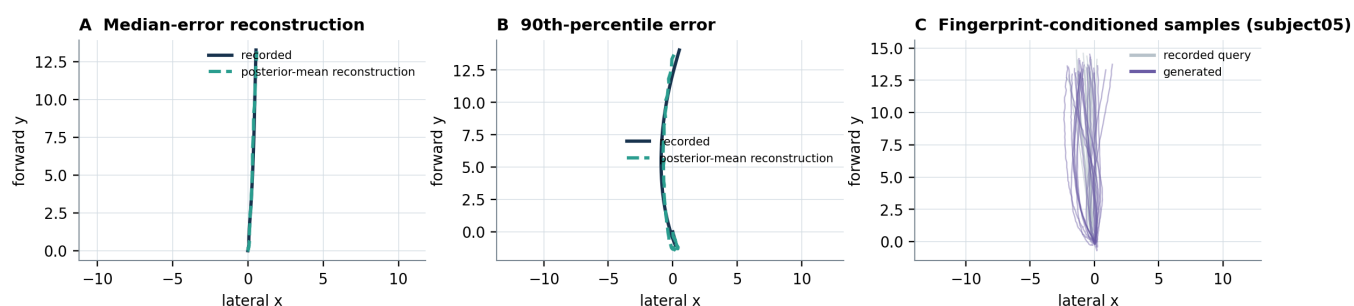


Figure A1. Panels A-B encode the observed trial and decode its posterior mean. Panel C does not encode the query trials; it samples around the participant's context fingerprint and compares the resulting family of trajectories with recorded query examples. One attractive plot cannot establish fidelity, which is why Tables 1-2 use every held-out participant.

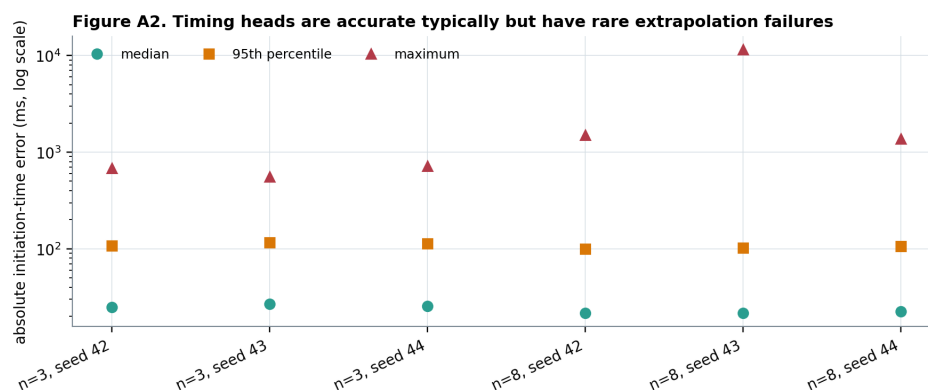


Figure A2. The median and 95th percentile are reasonable while maxima are orders of magnitude larger. This explains the apparently contradictory timing story: MAE can be around 35-45 ms while pooled R^2 is poor or negative. The model is usually close but occasionally catastrophically wrong.

Questions you should be able to answer

Question	Short accurate answer
Why target onset rather than finger onset?	It preserves waiting and initiation strategy; physical initiation time remains an explicit output.
Why resample to 100 phases?	The network needs fixed length. Because phase removes duration, timing is predicted separately.
Why average latent codes?	Context trials are repeated noisy observations in one learned coordinate system; their mean estimates a participant center.
Why is spline reconstruction better?	A low-rank linear shape basis is highly efficient for these smooth trajectories; CVAE also learns stochastic generation and timing.
Why use CVAE if U-VAE is similar?	It matches the desired task-conditional interface, but its incremental conditioning value remains unproven and is reported honestly.
Do components prove strategy?	No. They are model-dependent kinematic primitives that may support later strategy interpretation.
What is the main positive result?	Participant fingerprints support above-chance enrollment; $n=8$ generates timing/features better than spline and task-only Ridge.
What is the main limitation?	The requested $n=3$ code does not establish full distribution benefit over task metadata, and conditioning is not validated.

Claims to avoid

Avoid: 'CVAE is more accurate'; ' $n=3$ predicts the subject'; 'latent z_1 controls peak count'; 'component count is the participant's strategy'; 'the data are insufficient' without a learning-curve test; or ' R^2 =negative means the model learned nothing.' Replace each with endpoint-specific, test-backed wording from this guide.

Final interpretation

The project succeeds as a deep-learning benchmark and analysis: it establishes a leakage-resistant participant-fingerprint task, shows where a CVAE adds value, identifies where a classical spline remains stronger, and exposes two unresolved scientific issues - conditioning validity and submovement model order. It does not yet prove a fully controllable 2-3D biological strategy space.